

Waypoint vs. Gauss-Markov Mobility Comparison: (Phase 2 Item 1)

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- **Distinct scenario/speed configurations evaluated: 12**

Experimental design overview

Each configuration corresponds to one simulation run comparing the new Waypoint + dwell-time mobility model (patrol / transport / work robot archetypes, now with real 3D height variation) against the existing GaussMarkovMobilityModel baseline, evaluating whether structured robot movement changes WiFi-cellular switching behavior versus a random walk.

Design element	Specification
Total configurations	12 simulation runs
Scenarios	gaussmarkov_baseline, patrol, transport, work
Speeds swept	0.5, 2.0, 5.0 m/s (waypoint); 2.0 m/s (baseline head-to-head)
RNG seeds	6 (project convention reserves 6, 7, 8 for 3-repetition averages)
Parameters held constant	STA count 5; simulation duration 90s; RSSI handover threshold -80 dBm

1. Head-to-Head Comparison @ 2.0 m/s

This section presents the top-level switching and signal-quality indicators for each mobility scenario at the shared baseline speed, before the per-speed breakdown in Section 2.

Section Summary

- **Best reliability:** patrol reaches **97.95 ± 0.00%** packet-weighted PDR.

- **Fastest switch recovery:** `transport` averages **87.8 ± 0.0 ms** service interruption.
- **Slowest switch recovery:** `work` averages **637.4 ± 0.0 ms** service interruption.
- **Gauss-Markov RSSI abruptness:** p95 $|\Delta \text{RSSI}| = \mathbf{1.61 \pm 0.00 \text{ dB}}$ — the reference point Section 3 checks against.

Scenario	Runs	Switch events / 100s	Burstiness (CoV)	Mean $ \Delta \text{RSSI} $ (dB)	p95 $ \Delta \text{RSSI} $ (dB)	Mean interruption (ms)	Packet-weighted PDR (%)
gaussmarkov_baseline	1	34.44 ± 0.00	0.83 ± 0.00	0.35 ± 0.00	1.61 ± 0.00	284.7 ± 0.0	95.84 ± 0.00
patrol	1	35.56 ± 0.00	0.84 ± 0.00	0.21 ± 0.00	0.23 ± 0.00	194.3 ± 0.0	97.95 ± 0.00
transport	1	40.00 ± 0.00	0.81 ± 0.00	0.36 ± 0.00	1.11 ± 0.00	87.8 ± 0.0	97.84 ± 0.00
work	1	46.67 ± 0.00	0.56 ± 0.00	0.21 ± 0.00	0.73 ± 0.00	637.4 ± 0.0	88.51 ± 0.00

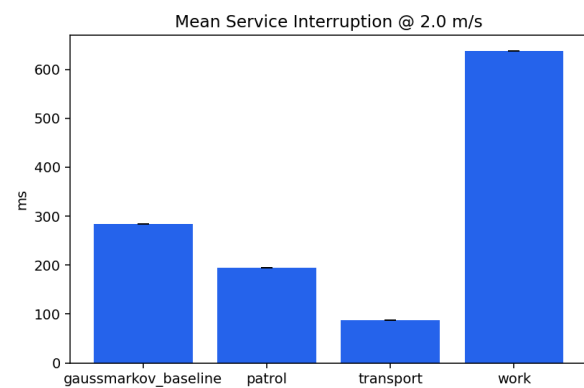
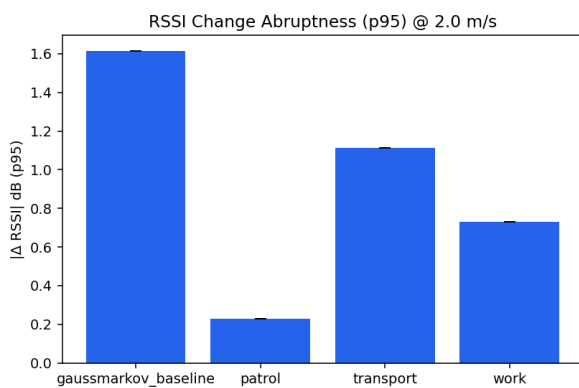
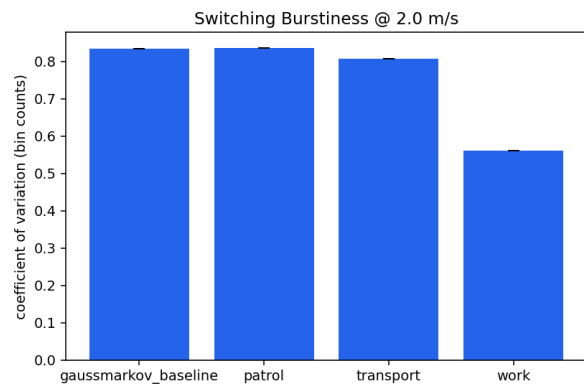
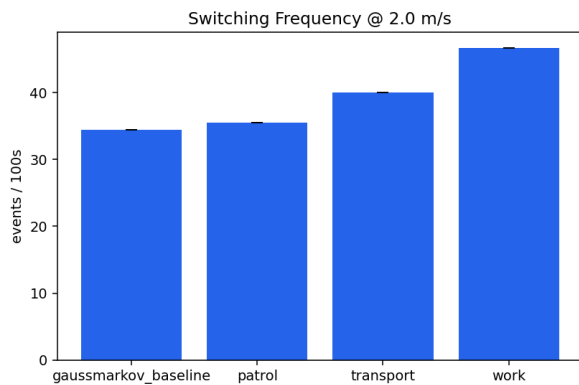


Figure: Switching frequency, burstiness, RSSI-change abruptness, and mean service interruption across scenarios at 2.0 m/s.

2. Waypoint Speed Sensitivity (0.5 / 2.0 / 5.0 m/s)

This section isolates robot speed as the variable of interest, per the enhancement plan's 3-level speed comparison (section 2.3): 0.5 m/s (precision work), 2.0 m/s (movement), 5.0 m/s (existing baseline).

Section Summary

- patrol from **0.5** to **5.0 m/s**: p95 $|\Delta\text{RSSI}|$ 0.07 $\rightarrow$ 0.59 dB (+0.52 dB).
- transport from **0.5** to **5.0 m/s**: p95 $|\Delta\text{RSSI}|$ 0.40 $\rightarrow$ 2.62 dB (+2.22 dB).
- work from **0.5** to **5.0 m/s**: p95 $|\Delta\text{RSSI}|$ 0.09 $\rightarrow$ 1.64 dB (+1.54 dB).

Scenario	Speed (m/s)	Runs	Switch events / 100s	Burstiness (CoV)	Mean $ \Delta\text{RSSI} $ (dB)	p95 $ \Delta\text{RSSI} $ (dB)
patrol	0.5	1	46.67 $\pm$ 0.00	0.67 $\pm$ 0.00	0.15 $\pm$ 0.00	0.07 $\pm$ 0.00
patrol	2.0	1	35.56 $\pm$ 0.00	0.84 $\pm$ 0.00	0.21 $\pm$ 0.00	0.23 $\pm$ 0.00
patrol	5.0	1	18.89 $\pm$ 0.00	1.69 $\pm$ 0.00	0.31 $\pm$ 0.00	0.59 $\pm$ 0.00
transport	0.5	1	42.22 $\pm$ 0.00	0.72 $\pm$ 0.00	0.19 $\pm$ 0.00	0.40 $\pm$ 0.00
transport	2.0	1	40.00 $\pm$ 0.00	0.81 $\pm$ 0.00	0.36 $\pm$ 0.00	1.11 $\pm$ 0.00
transport	5.0	1	32.22 $\pm$ 0.00	0.99 $\pm$ 0.00	0.70 $\pm$ 0.00	2.62 $\pm$ 0.00
work	0.5	1	40.00 $\pm$ 0.00	0.79 $\pm$ 0.00	0.12 $\pm$ 0.00	0.09 $\pm$ 0.00
work	2.0	1	46.67 $\pm$ 0.00	0.56 $\pm$ 0.00	0.21 $\pm$ 0.00	0.73 $\pm$ 0.00
work	5.0	1	35.56 $\pm$ 0.00	0.92 $\pm$ 0.00	0.43 $\pm$ 0.00	1.64 $\pm$ 0.00

Figure interpretation: RSSI-change abruptness increases with speed across all three robot types — the clearest, most physically-expected result in this study. Faster movement covers more distance between fixed-interval samples, so signal strength changes more per second.

3. Verdict Against the Enhancement Plan's Predictions

The enhancement plan predicts two effects of switching from Gauss-Markov to Waypoint mobility: switching events should become more concentrated (vs. dispersed), and RSSI variation should become more abrupt in specific zones (vs. gradual).

Section Summary

- **Switching events dispersed → concentrated:** NOT CONFIRMED — Gauss-Markov burstiness (CoV) = **0.83**, waypoint average = **0.73**.
- **RSSI variation gradual → abrupt-change zones:** NOT CONFIRMED — Gauss-Markov p95 $|\Delta\text{RSSI}|$ = **1.61 dB**, waypoint average = **0.69 dB**.